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MID SEMESTER EXAMINATION SEPTEMBER - 2025

Name of the Program: **B.Tech., ME**

Semester: **Vth**

Name of the Course: **Mechanical Measurements and Metrology**

Course Code: **(TME 510)**

Time: 90 Minutes

Maximum Marks: 50

Note:-

- (i) Answer **all the questions** by choosing **anyone of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10marks)	CO1 and CO2
(a)	Differentiate between line standards and end standards. Give at least three differences with suitable examples.	
OR		
(b)	Define tolerance. Explain its importance in manufacturing. Discuss how tolerance specifications influence assembly, product quality, and cost of manufacturing.	
Q2	(10 marks)	CO1 and CO2
(a)	What are slip gauges? Describe how slip gauges are manufactured and mention their standard materials, grades, and accuracy.	
OR		
(b)	A shaft is specified as 25h6. The fundamental deviation for h shaft is zero. If the IT6 tolerance for 25 mm is 16 μm , find: - Upper and lower deviations of the shaft - Maximum and minimum size of the shaft	
Q3	(10 marks)	CO1 and CO2
(a)	Describe the subdivision of standards into primary, secondary, tertiary, and working standards with examples.	
OR		
(b)	Define geometrical tolerances and explain different types (form, orientation, location, runout) with neat sketches.	
Q4	(10 marks)	CO1 and CO2
(a)	Calibration of End Bar with Temperature Correction Given: Nominal Length = 60.00 mm, Comparator Reading = -0.005 mm Calibration temperature = $15\text{ }^{\circ}\text{C}$, Standard temperature = $20\text{ }^{\circ}\text{C}$ Coefficient of thermal expansion (α) = $11 \times 10^{-6}\text{ }/^{\circ}\text{C}$ Find the actual length of the end bar after temperature correction.	
OR		
(b)	Discuss the classification of gauges and give examples for limit gauges, plug gauges, snap gauges, and ring gauges.	
Q5	(10 marks)	CO1 and CO2
(a)	Building of Slip Gauges: Build a length of 69.48 mm using the minimum number of slip gauges from a metric M-87 set. (Note: Show selection logic from 1.0 mm, 1.005 mm, 1.01–1.49 mm (0.01 mm step), 1.5–24.5 mm (0.5 mm step), 25–100 mm (1 mm step)).	
OR		
(b)	What Write short notes on: - Fundamental deviation - Basic size - Zero line - ISO system of limits and fits	

